

Semantic Refined Prompting based Automated Essay Scoring System

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Abstract—The usage of Large Language Models (LLMs) has made significant contributions in various areas including essay scoring. However, its performance is not comparable to human grades due to the reasons that the auto-generated scores are not done based on semantic understanding. Existing research on this topic focuses on grading with predefined rubrics. Since grading can't be relied only on the predefined rubrics in this paper, we propose an LLM-based scoring mechanism that considers the predefined and semantic refined rubrics. Additionally, the scores obtained are evaluated for consistency and fairness. This system is evaluated using the ASAP dataset and the outcomes demonstrate the effectiveness of the approach in terms of RMSE and MAP.

Keywords: Automated Essay Scoring (AES), prompt engineering, Large Language Models (LLM).

I. INTRODUCTION

Essay writing is an indispensable part of every individual, as it improves one's creativity, critical thinking, research skills, etc. Thus, it plays an irreplaceable role in academics. Every year thousands of students take standardized exams and end up writing thousands of essays on the same topics. Grading these essays manually is highly laborious as they are subjective in nature. Thus, automated essay grading can be highly beneficial as it reduces human error and biasing. With an automated system, teachers can use their time to optimize the educational content.

Automatic essay scoring holds a lot of vibrancy in terms of research, as there exists a huge gap between human grades and auto grades. To fill this gap, automated essay scoring remains a topic of interest in both linguistics and machine learning. Large Language Models (LLMs) [1-3], with their immense knowledge, have shown excellence in multiple Natural Language Processing tasks like sentiment analysis [4] and other tasks that require logical reasoning. To enhance the grading process of the students, the LLMs are used specifically in academics as a question-and-answering system [5, 6]. Having found its widespread usage, it is evident that the LLMs can be used for evaluating the student's essays accurately and fairly. However, teacher-based grading is specific to essays as it requires a meticulous checking of each student's answers. Human grading relies upon the standard rubrics specific to questions and is adjusted according to the individual student responses. With the rubrics in hand, humans can provide better scores, ensuring fairness. LLMs can be utilized for grading like a human by incorporating

student responses into the rubric design with a prompting mechanism. On the other hand, an automated scoring system to achieve good detection accuracy requires them to understand the essays semantically, and thus they can grade like a human. In this context, active learning-based systems have shown significant contributions in terms of detection accuracy and computation speed [7].

Thus, if the prompting is enhanced with semantic refinement, it can aid the LLM to achieve a score comparable to a human. In this work, a BERT-based prompt model is integrated with semantic refinement for better essay scoring. The model proposed is novel and innovative, as demonstrated by the following points:

- A unique prompt including the scoring keyword slot is designed within the existing prompt template, ensuring the scoring becomes human-like and the semantic-based prompts get completed.
- The model's understanding is strengthened by the semantic refinement layer that uses stacked multi-layer convolutions with gating.
- The gating mechanism ensures the combination of the pooling vector with the tokens that ensures better scoring.
- The semantic extraction here is carried out using a multigrain phrase extraction that works in conjunction with the semantic refinement layer.

Thus, the major contributions of this work are as follows:

- A prompt model proposed explores the knowledge related to the automated scoring by the pre-trained BERT model, and thus the input representation becomes close to the auto scoring.
- The model's understanding of prompts is further enhanced with the improved semantic refinement using the multi-semantic composition method for better scoring.

The rest of the paper is organized as follows:

Section 2 presents the literature survey, followed by the modified, semantically refined BERT model in section 3. Section 4 presents the experimentation and result analysis followed by the concluding remarks in section 5.

II. LITERATURE SURVEY

Automated Essay Scoring (AES) holds a significant position by its intersection with Natural Language Processing (NLP) and the education sector. The earlier days of AES were carried out as a two-stage process [8] in which the first part of the scoring involves the feature extraction [9] followed by the latent vector representations. The subsequent step involved the usage of the baseline machine learning-based classification and regression models that were used for the essay scoring [10]. Deep learning models' advancements in prediction have shown comparable performance in various domains and have witnessed their integration in essay scoring. Convolutional neural networks (CNNs), long short-term memory networks (LSTMs), attention-based models, and other deep learning techniques have been used to improve the score predictions [11]. However, using conventional approaches results in more training, and the computation overhead is also high.

The next evolution of AES was in the usage of the pre-trained language models for further enhancing the scores, as well as the higher cost associated with training and processing can be reduced. Bidirectional encoders from transformers were utilized by [12, 13] for the automatic assessment of the essays that comprised short answers. The transformer-based approaches used BERT as the model, and several other works aimed at improving their performance by fine-tuning the BERT models as mentioned in [14, 15]. The fine-tuning is carried out with the modification of the loss function as in [14], where the loss is considered as a combination of the regression and ranking. The BERT scoring is scaled to multi-scale essays in [15] with fine-tuning.

During recent days, the large language models (LLMs) have played a pivotal role in AES [16, 17]. GPT-4 is used for grading the programming assignments, and the suitability of using the LLMs for grading the code-based responses is verified in [18], and the overall accuracy obtained is 75%. Advanced prompting techniques are used to improve the scores of AES. One-shot prompting, along with assistance from the text similarity model, is used in [19] for obtaining the analytic score along with the holistic score and achieved a kappa score of 0.67 and 0.71 with its domain adaptation capability. The K-12 science assessments are validated in [20] using the GPT-4 with the Chain of Thought (CoT) prompts. The explanation for the scores is also obtained using the human loop approach. Using zero prompting, a comparative evaluation has been carried out in [21] using the FlanT5 and Llama2-chat prompting. It has also been reported that the LLMs exhibit positional biases while performing pairwise comparisons. Human-refined ideology is presented as self-refine in [22], where the output generated from the prompts is iteratively refined using the feedback mechanism. This methodology of refinement is done without any training data, and the LLMs are optimized like human optimization. ALLURE [23] performs the comparison of the LLM-generated output to that of the manual annotations, and then the deviations are observed and fed back to the evaluator, significantly improving the performance of the LLM, and thus the need for training is removed and the dependency for manual annotation is reduced in this approach. The problem of generalizability in AES is addressed using the zero-shot, few-shot, and p-tuning methods adopted in [24], in which the

comment with the score is fed as a prompt, proving the usability of LLMs in improving the score of the essays. While the usage of LLMs has significantly improved the scoring of essays, its usability in real-world scenarios is not comparable to the human [25]. The coherency difference between the LLM and human is studied in [26] and the biases among them. Overall, the major difference between human graders and LLMs happens due to the contextual variation. In contrast to the various approaches used here, the approach adopted here adopts a methodology for the grading process that comprises the rubric generation based on the semantic refinement followed by the grading. Multiple agents are used for the generation, refinement, and grading process that aids in obtaining consistent results.

III. PROPOSED SEMANTIC REFINED PROMPTING FOR AUTOMATED ESSAY SCORING

Semantic refined prompting comprises multiple agents that aid in enhancing the overall performance of the LLMs that are specific for the essay scoring. The overall architecture is presented in Figure 1. As shown in Figure, the grading system involves three steps Semantic Refined Rubric, Prompt based grading, and Score validator. In the Semantic Refined Rubric step, the scoring criteria is modified to include the semantic components which in turn returns a score that closely resembles the human scores. Using these semantic rubrics, the prompts obtain the grading for the essay set. The Score validator step validates the score's outcome for any undesirable score. In the following section, a detailed explanation of each of the stages is provided.

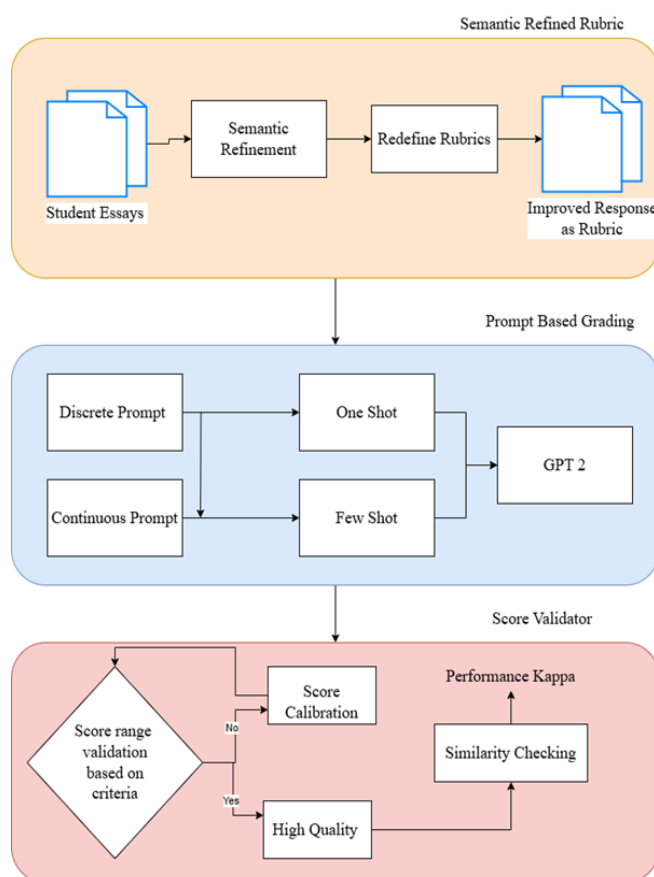


Figure 1: Overall architecture of the Semantic Refined Prompting

A. Semantic Refined Rubric Generation:

The overall process adopted for rubric generation is outlined here. This is the initial step that facilitates the grading task. Human grading is better than automated grading because the individual student answers and the semantic consideration aid the human scorer to score well, which breaks the monotony followed by the computerized grading system. So, the main purpose of this study is to customize the rubric so that the system performs the grading based on the semantics available with individual student responses. The general idea behind the rubric generation is to guide the LLMs about the grading process, and it is generated as an iterative process. The novelty that is specific to this work lies with the student sample selection and the scoring, as here in our approach, instead of relying on the human evaluators for the samples, we extract the semantics from the essays using the Sentence-BERT LLMs and feed them into a pre-trained S-LLM. Through this approach, the semantic and the linguistic features are integrated, which can improve the LLMs grading performance. The iterations generate the rubrics considering the predefined human rubrics as well as the rubrics from the Sentence-BERT. This way the LLMs get knowledge not only about the semantics behind the essays but also the way to grade.

In this manner, the rubrics are optimized and this optimized rubric is then used as an input for the next iteration. The rubrics that are generated after the execution of all the iterations is the final rubric. Figure 2 presents the prompt for the semantic refined Rubric generation.

"Computers have revolutionized how we work and communicate.

They improve efficiency but can also lead to issues like lack of exercise.

This essay explores the benefits and challenges.",

"People today spend a lot of time using computers.

While they connect us globally, they also reduce the time we spend with family and friends. I will discuss both sides in this essay."

The current question is: {question}.

The initial grading rubric is: {human_rubric} and {semantic_rubric}.

The full_points is {full_points}.

The samples of correctly grading are: {sample_studata,SBert sentencedata}

Your generated grading criteria should illustrate the key points of the answer for each question or sub-question and the rules for assigning points.

Figure 2: Example of prompts for Semantic rubric generation

Since the rubric generation is driven by the samples taken from the students' responses and the semantics obtained from the sentence BERT the performance of the rubric is sample-dependent. To enhance the reliability and validity of our approach, we employed a random sampling technique for selecting these student responses. This method ensures that the samples are representative of a larger population, allowing the rubric to perform consistently and accurately across diverse student inputs. By implementing random sampling, we seek to minimize potential biases that could arise from using a non-representative set of responses, ultimately leading to a more robust and effective assessment tool.

B. Prompt Based Grading:

This section details the prompts that are used for the grading. The evaluation score for the essay is generated based on the student responses, the questions, and the semantic refined rubrics are framed as a prompt and given as input to the LLM model for performing the grading process. The prompts if written appropriately have a greater impact on the grades and thus in this work, two prompting strategies are adopted. The details of the prompting strategy are as follows:

- **One-shot Prompts:** Since this involves essay scoring, giving a template for in-context learning can improve the LLM capability to a greater extent. Thus here, the additional template for the LLM is given as a prompt.
- **Few shot Prompts:** Since here we use semantic refinement along with the student responses, thus to provide detailed guidance this prompting is also used. The few short prompts are combined along with the one-shot to produce the final scores.

C. Score Validator

One of the common concerns with the LLMs is the issues with hallucination. Thus, validating the score of the LLMs is crucial for the identification of the scores that don't hold reasonable reasons. The scored essays can either be validated individually or with a group. However, the validation criteria remain the same. The criteria attempt to match the scores of the essays after the grading to that of the rubric values and any value that is beyond the level mentioned by the rubric is considered an outlier and is reported as an anomaly. After the anomaly detection, if the score falls within the desired then the kappa score corresponding to that essay set is validated to ensure the accuracy of the auto-generated scores.

IV. EXPERIMENTATION AND RESULT ANALYSIS

A. Dataset

The proposed system is implemented and tested using the Automated Student Assessment Prize (ASAP) dataset [27]. This is one of the most popular AES datasets introduced by the Flora Hewlett Foundation in 2012 for a Kaggle competition. The dataset comprises a total of 12,978 essays that comprised narrative and source-dependent responses. The essay size is varied in terms of size, style, and content. These essays are human-graded with varied criteria set for

essays based on the content. 8 different essay sets are available with varied criteria for scoring.

B. Experimental Setup:

The entire implementation of this system is using Python language on a server with Nvidia GPU and 400 GB memory. The open source LLM model GPT2 and Mistral are employed for the implementation of the rubric and prompts.

C. Performance Evaluation of the Rubrics:

The rubrics that combine the semantic refinement and its impact in improving the essay score is evaluated in this section. Table 1 presents the evaluation results of the ASAP dataset.

Table 1: Performance of different prompt strategies on grading performance for the ASAP dataset.

Method	MAE	RMSE	Pearson	MAE	RMSE	Pearson
Essay Set 1				Essay Set 2		
Bert	4.12	0.21	0.70	8.90	0.59	0.4
One-Shot	3.54	0.29	0.67	5.76	0.43	0.35
Few-shot	3.92	0.24	0.76	5.32	0.46	0.53
Combined	3.32	0.21	0.79	4.32	0.32	0.41

As observed from Table 1, the evaluation results on the ASAP dataset is observed for 4 different prompting. Among the four techniques, the joint prompt considering the fine details of the essay performed better than the other methodologies.

V. CONCLUSION

This work presents a semantic refined automatic scoring system for the essays. Current scoring systems face challenges in terms of the accuracy of the score and fairness as they lack human understanding. To tackle this a scoring system that closely resembles the human grading evaluation is suggested and evaluated. The effectiveness of the proposed rubric is evaluated using the ASAP dataset showing comparable performance. However, the method proposed score is domain-specific, and thus the system usability on diversified domains is left as a future scope.

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